

## homework<sub>1</sub>

June 12, 2026

1. If  $R$  is a ring, show that  $M_n(R)$  is a ring under the usual operations of matrix addition and multiplication.
2. Let  $(G, +)$  be an Abelian group. An **endomorphism** of  $G$  is a group homomorphism  $\varphi : G \rightarrow G$ . Let  $\text{End}(G)$  denote the set of all endomorphisms on  $G$ . Given  $\varphi, \psi \in \text{End}(G)$ , define

$$(\varphi + \psi)(g) := \varphi(g) + \psi(g), \quad \text{and} \quad (\varphi \times \psi)(g) := \varphi(\psi(g)).$$

- (a) Show that  $+$  and  $\times$  are well-defined binary operations on  $\text{End}(G)$ .
  - (b) Show that  $(\text{End}(G), +, \times)$  is a ring.
3. Let  $R$  be a ring and  $a, b \in R$ . Show that
    - (a)  $0 \cdot a = a \cdot 0 = 0$ .
    - (b) For any  $n \in \mathbb{Z}$ ,  $(na)b = n(ab) = a(nb)$ . [Hint: Use Induction.]
    - (c) If  $R$  is unital, then the multiplicative identity is unique. Moreover,  $(-1) \cdot a = (-a)$  for all  $a \in R$ .
  4. Let  $R$  be a commutative ring and let  $S$  and  $T$  be two subsets of  $R$ . Define

$$ST := \bigcup_{n=1}^{\infty} \left\{ \sum_{i=1}^n s_i t_i : s_i \in S, t_i \in T \right\}.$$

(In other words, elements in  $ST$  are finite sums of the form  $\sum_{i=1}^n s_i t_i$ , where  $s_i \in S$  and  $t_i \in T$ , while the number  $n$  may vary.)

- (a) If  $S$  and  $T$  are both subrings of  $R$ , prove that  $ST$  is a subring.
  - (b) If  $S$  and  $T$  are both ideals, then prove that  $ST$  is an ideal.
5. Fix  $m, n \in \mathbb{Z}$  and let  $d := \gcd(m, n)$ . If  $I$  denotes the ideal generated by  $\{m, n\}$ , then show that  $I = (d)$ , the principal ideal generated by  $d$  (See Definition 5.3).

### Extra Problems

You need not submit these problems; they are here only for practice.

**\*\*Definition:\*\*** A function  $f : \mathbb{R} \rightarrow \mathbb{R}$  is said to have **\*\*compact support\*\*** if there exists  $a, b \in \mathbb{R}$  such that  $f(x) = 0$  for all  $x \notin [a, b]$ .

Define

$$C_c(\mathbb{R}) := \{f : \mathbb{R} \rightarrow \mathbb{C} : f \text{ is continuous and has compact support}\}.$$

Equip  $C_c(\mathbb{R})$  with the pointwise operations as in Part (v) of Example 1.2 in the notes. You need not check that  $C_c(\mathbb{R})$  is a ring (this is easy to do).

1. Prove that  $C_c(\mathbb{R})$  is not unital.

2. Let  $R$  be a ring,  $S$  be a subring of  $R$  and let  $I$  be an ideal of  $R$ .
- (a) Prove that  $S + I := \{s + a : s \in S, a \in I\}$  is a subring of  $R$ .
  - (b) Prove that  $I \triangleleft S + I$ .
  - (c) Prove that  $S \cap I \triangleleft S$ .
  - (d) Prove that

$$\frac{S + I}{I} \cong \frac{S}{S \cap I}$$

[Hint: Define  $\varphi : S \rightarrow (S + I)/I$  by  $s \mapsto s + I$ .]

3. Let  $D \in \mathbb{Q}$  be a rational number that is not a perfect square. Define

$$R := \{a + b\sqrt{D} : a, b \in \mathbb{Q}\}.$$

- (a) Show that  $R$  is a subring of  $\mathbb{C}$ .
  - (b) If  $(a + b\sqrt{D}) \neq 0$ , then show that  $a^2 - b^2D \neq 0$ .
  - (c) Use Part (b) to prove that  $R$  is a field. This field is denoted  $\mathbb{Q}(\sqrt{D})$ , and is called a **quadratic field**.
4. For a unital ring  $R$ , define the **center** of  $R$  to be

$$\mathcal{Z}(R) := \{a \in R : ab = ba \text{ for all } b \in R\}.$$

Show that  $\mathcal{Z}(R)$  is a subring of  $R$ .

Given:  $R$  is a ring.

We show that  $M_n(\mathbb{C}R)$  is a ring under  $+$ , matrix multiplication.

*Proof.* (iii) Fix  $n \in \mathbb{N}$ . Show  $M_n(\mathbb{C}R)$  is a ring under addition and multiplication of real numbers is  $C = AB$  s.t.

$$c_{ij} = \sum_{k=1}^n a_{ik}b_{kj} \quad (\text{Check})$$

**Associativity of multiplication:**

$$A \cdot (B \cdot C) = A \cdot D \quad \text{where} \quad d_{ij} = \sum_{k=1}^n b_{ik}c_{kj}$$

$$\Rightarrow A \cdot (B \cdot C) = \left( \sum_{k=1}^n a_{ik}d_{kj} \right)_{n \times n}$$

$$= \left( \sum_{k=1}^n a_{ik} \left( \sum_{l=1}^n b_{il}c_{lj} \right) \right)_{n \times n}$$

$$= \left( \sum_{l=1}^n \left( \sum_{k=1}^n a_{ik}b_{kl} \right) c_{lj} \right)_{n \times n}$$

$$= (A \cdot B) \cdot C$$

**Distributive Property:**

$$\begin{aligned} AB + AC &= \left( \sum_{k=1}^n a_{ik}b_{kj} + \sum_{k=1}^n a_{ik}c_{kj} \right)_{n \times n} \\ &= \left( \sum_{k=1}^n a_{ik}(b_{kj} + c_{kj}) \right)_{n \times n} \\ &= A(B + C) \end{aligned}$$

Similarly, we can show  $(A + B)C = AC + BC$ .

(iv) Conclude that if  $R$  is a ring, then  $M_n(R)$  is a ring.

□

Given  $(G, +)$  is an Abelian group,

$$\text{End}(G) = \{\varphi : G \rightarrow G \mid \varphi \text{ is a group homomorphism}\}$$

For  $\varphi, \psi \in \text{End}(G)$ ,

$$\begin{aligned} (\varphi + \psi)(g) &= \varphi(g) + \psi(g) \\ (\varphi \times \psi)(g) &= \varphi(\psi(g)) \end{aligned}$$

(a) Show the operations given above are well defined.

*Proof.* We show  $\varphi + \psi \in \text{End}(G)$ . Consider  $g, h \in G$ :

$$(\varphi + \psi)(g + h) = \varphi(g + h) + \psi(g + h) = \varphi(g) + \varphi(h) + \psi(g) + \psi(h)$$

Using the abelian property here,

$$= \varphi(g) + \psi(g) + \varphi(h) + \psi(h) = (\varphi + \psi)(g) + (\varphi + \psi)(h)$$

Thus,  $\varphi + \psi \in \text{End}(G)$ .

We show  $\varphi \times \psi \in \text{End}(G)$ . Consider  $g, h \in G$ :

$$\begin{aligned} (\varphi \times \psi)(g + h) &= \varphi(\psi(g + h)) = \varphi(\psi(g) + \psi(h)) = \varphi(\psi(g)) + \varphi(\psi(h)) \\ &= (\varphi \times \psi)(g) + (\varphi \times \psi)(h) \end{aligned}$$

Thus,  $\varphi \times \psi \in \text{End}(G)$ .

□

(b) Prove  $(\text{End}(G), +, \times)$  is a ring.

*Proof.* As  $G$  is an abelian group,  $(\text{End}(G), +)$  is an abelian group.

Associativity holds as composition of functions is associative.

Distributive Property:

$$\begin{aligned}\varphi \times (\psi_1 + \psi_2)(g) &= \varphi \times (\psi_1 + \psi_2)(g) \\ &= \varphi(\psi_1(g) + \psi_2(g)) = \varphi(\psi_1(g)) + \varphi(\psi_2(g)) \\ &= (\varphi \times \psi_1)(g) + (\varphi \times \psi_2)(g) \\ &= (\varphi \times \psi_1 + \varphi \times \psi_2)(g)\end{aligned}$$

Thus,  $(\text{End}(G), +, \times)$  is a ring. □

$$(\varphi_1 + \varphi_2) \times \varphi(g) = (\varphi_1 + \varphi_2)(\varphi(g)) = \varphi_1(\varphi(g)) + \varphi_2(\varphi(g)) = \varphi_1 \times \varphi(g) + \varphi_2 \times \varphi(g).$$

1. Given:  $(R, +, \cdot)$  is a ring;  $a, b \in R$

a) Show:  $O \cdot a = a \cdot O = O$

$$\text{Proof. } O \cdot a = (a - a) \cdot a = a^2 - a^2 = O$$

$$a \cdot O = a \cdot (a - a) = a^2 - a^2 = O$$

□

b) Show:  $(na)b = n(ab) = a(nb)$

*Proof.* Base case ( $n = 1$ ): Trivial.

Induction Hypothesis:  $(na)b = n(ab) = a(nb)$

$$(a + a + \cdots + a \text{ (n+1 times)})b = (na + a)b = nab + ab = (n + 1)ab$$

$$a(n + 1 \text{ times } b) = a(n \text{ times } b + b) = nab + ab = (n + 1)ab$$

□

c) If  $R$  is unital, prove: 1 is multiplicative identity is unique.

*Proof.*  $1_A = 1_A \cdot 1_B = 1_B \implies$  Multiplicative identity is unique.

Show:  $(-1) \cdot a = -a$  for any  $a \in R$

$$\text{Proof: } a + (-1)a = 1 \cdot a + (-1)a = (1 + (-1))a = 0$$

$$\implies (-1)a \text{ is the additive inverse } \implies (-1) \cdot a = -a$$

□

2. Given:  $R$  is commutative ring,  $S, T \subseteq R$

$$ST = \bigcup_{n=1}^{\infty} \{ \sum_{i=1}^n s_i t_i \mid s_i \in S, t_i \in T \}$$

a) If  $S$  and  $T$  are both subrings, show:  $ST$  is a subring.

*Proof.* Assume  $S$  and  $T$  to be non-empty. □

Let  $a, b \in ST$  such that  $a = \sum_{i=1}^{n_1} s_i t_i$  and  $b = \sum_{i=1}^{n_2} s'_i t'_i$ .

Without loss of generality, assume  $n_2 > n_1$ . Then,

$$b - a = \sum_{i=1}^{n_2} s'_i t'_i - \sum_{i=1}^{n_1} s_i t_i = \sum_{i=n_1+1}^{n_2} s'_i t'_i$$

by the Abelian property.

$$\sum_{i=n_1+1}^{n_2} s'_i t'_i = \sum_{i=1}^{n_2} s_i t_i$$

where  $i = 1$  to  $i = n_1 = 0$  for the first entry.

Thus,  $(ST, +) \prec (R, +)$ .

*Proof.* Check the ring property:

$$\left( \sum_{i=1}^{n_1} s_i t_i \right) \left( \sum_{j=1}^{n_2} s'_j t'_j \right)$$

Without loss of generality, assume  $n_1 < n_2$ . Now set all terms to zero in the first entry after  $i > n_1$ ,

$$\left( \sum_{i=1}^{n_2} s_i t_i \right) \left( \sum_{j=1}^{n_2} s'_j t'_j \right)$$

Every term will be of the form  $s_i s'_j t_i t'_j$  for some  $1 \leq i, j \leq n_2$  and as  $R$  is a commutative ring, we get  $s_i s'_j t_i t'_j$  and as  $S, T$  are subrings,  $s_i s'_j \in S$  and  $t_i t'_j \in T$  respectively. □

**Definition 0.1.** If  $S, T$  are both ideals, then  $ST$  is an ideal.

*Proof.* Consider  $\sum_{i=1}^n s_i t_i \in ST$  and  $r \in R$ . Then,

$$r \left( \sum_{i=1}^n s_i t_i \right) = \sum_{i=1}^n (r s_i) t_i$$

as  $S$  is an ideal, we get  $r s_i \in S \forall i \leq n$ .

$$\left( \sum_{i=1}^n s_i t_i \right) r = \sum_{i=1}^n s_i (t_i r)$$

as  $T$  is an ideal, we get  $t_i r \in T \forall i \leq n$ . □

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Given  $m, n \in \mathbb{Z}$  and  $d = \gcd(m, n)$ , let  $I$  be the ideal generated by  $\{m, n\}$ .

**Theorem 0.1.** Show that  $I = (d)$ .

*Proof.* Let  $X = \{J \subseteq \mathbb{Z} \mid J \text{ is an ideal and } X \subseteq J\}$ . Define

$$(X) = \bigcap_{J \in \mathcal{F}} J,$$

where  $\mathcal{F} = \{J \subseteq \mathbb{Z} \mid X \subseteq J\}$ .

If  $a \in (X)$ , then  $a \in J$  for every  $J \in \mathcal{F}$ .

Since  $J$  is an ideal of  $\mathbb{Z}$ , it must also be a subgroup of  $\mathbb{Z}$ . Thus,  $\exists n_J \in \mathbb{Z}$  such that  $J = n_J \mathbb{Z}$ .

If  $x \in J$ , then  $n_J \mid x$  and  $n_J \mid y$  for some  $x, y \in X$ .

Since  $d = \gcd(x, y)$ , we have  $d \mid n_J$ .

Therefore,  $n_J \mathbb{Z} \subseteq d\mathbb{Z}$ , where  $n_J$  is the integer corresponding to  $J$ .

Hence,  $(X) \subseteq d\mathbb{Z}$ .

**Claim:**  $d\mathbb{Z}$  is the smallest ideal containing  $X$ .

Consider  $n\mathbb{Z} \subseteq \mathbb{Z}$  such that  $X \subseteq n\mathbb{Z}$  and  $n\mathbb{Z} \subseteq d\mathbb{Z}$ .

Let  $nk_1 = x$  and  $nk_2 = y$  for some  $k_1, k_2 \in \mathbb{Z}$ .

By Bézout's identity, there exist  $r_1, r_2 \in \mathbb{Z}$  such that

$$xr_1 + yr_2 = d.$$

Substituting  $x = nk_1$  and  $y = nk_2$ , we get

$$n(k_1r_1 + k_2r_2) = d.$$

Thus,  $n \mid d$ .

Hence,  $d\mathbb{Z} \subseteq n\mathbb{Z}$ , which implies  $n\mathbb{Z} = d\mathbb{Z}$ .

Therefore,  $I = (d)$ . □

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**Definition 0.2.** Given the set

$$C_c(\mathbb{R}) := \{f : \mathbb{R} \rightarrow \mathbb{C} \mid f \text{ is continuous and has compact support}\},$$

where compact support means  $\exists a, b \in \mathbb{R}$  such that

$$f(x) = 0 \text{ whenever } x \notin [a, b],$$

and  $(+, \times)$  are the same as usual addition and multiplication.

*Remark 0.1.* Observe that  $C_c(\mathbb{R})$  is a ring, and it is non-empty.

**Theorem 0.2.** Show that  $C_c(\mathbb{R})$  is not unital.

*Proof.* Assume there is  $g \in C_c(\mathbb{R})$  such that  $f \times g = f$  for any  $f \in C_c(\mathbb{R})$ .

If  $g(x_0) \neq 0$  and  $x_0 \in [a, b]$ , choose  $f \in C_c(\mathbb{R})$  such that

$$f(x) \neq 0 \text{ whenever } x \in [a', b']$$

where  $[a, b] \subset [a', b']$ .

Observe at  $x_0 = \frac{a'+a}{2}$ ,

$$g(x_0) = 0 \quad \text{and} \quad f(x_0) \neq 0 \implies f \times g(x_0) \neq 0.$$

This is a contradiction. Hence, no unit exists in  $C_c(\mathbb{R})$ . □

Given the content in the image, 2.

1. Given:  $R$  is a ring,  $S$  be a subring of  $R$  and  $I \trianglelefteq R$ .

(a)

**Theorem 0.3.** Prove:  $S + I := \{s + a \mid s \in S, a \in I\}$  is a subring.

*Proof.* Observe,  $S + I$  is non-empty. Then consider  $a', b' \in S + I$  where  $a' = s_1 + a$  and  $b' = s_2 + b$ . Now,

$$a' - b' = (s_1 + a) - (s_2 + b) = (s_1 - s_2) + (a - b) \in S + I$$

where  $s_1 - s_2 \in S$  and  $a - b \in I$ . Thus,  $(S + I, +) \leq (R, +)$ .

Now, let  $a', b' \in S + I$  where  $a' = s_1 + a$ ,  $b' = s_2 + b$ . Then,

$$a' \cdot b' = (s_1 + a)(s_2 + b) = s_1 s_2 + a s_2 + s_1 b + ab$$

where  $s_1 s_2 \in S$  and  $a s_2 + s_1 b + ab \in I$ . Therefore,  $(S + I, +, \cdot)$  is a subring of  $(R, +, \cdot)$ . □

(b)

**Theorem 0.4.** Prove:  $I \trianglelefteq S + I$ .

*Proof.* Consider  $a \in I$  and  $b' \in S + I$  such that  $b' = s + b$  where  $s \in S$ ,  $b \in I$ . Observe

$$ab = a(s + b) = as + ab,$$

as  $I \trianglelefteq R$ ,  $as \in I$  and  $ab \in I \Rightarrow ab \in I$ . Similarly,  $b'a \in I$ . □

(c)

**Theorem 0.5.** Show:  $S \cap I \trianglelefteq S$ .

*Proof.* As  $(I, +) \trianglelefteq R$ ,  $(S \cap I, +) \trianglelefteq S$  and  $0 \in S \cap I \Rightarrow S \cap I \neq \emptyset$ . If  $a \in S \cap I$  and  $s \in S$ , then  $a \in I$  and  $s \in S \Rightarrow as \in I$ . Now,  $a \in S$  and  $s \in S \Rightarrow as \in S$ .  $\square$

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Hence,  $as \in S \cap I \Rightarrow sa \in S \cap I$ .

$$\Rightarrow S \cap I \triangleleft S$$

*Proof.* (cd) Prove:  $\frac{S+I}{I} \cong \frac{S}{S \cap I}$

**Proof:**

$$1. \varphi : S \rightarrow \frac{S+I}{I} \quad (s \mapsto s + I)$$

**(1) Well definedness**

$$s_1 = s_2 \Rightarrow s_1 + I = s_2 + I$$

**(2)  $\varphi$  is a ring homomorphism**

$$\varphi(s + s') = s + s' + I = (s + I) + (s' + I) = \varphi(s) + \varphi(s')$$

$$\varphi(ss') = ss' + I = (s + I)(s' + I) = \varphi(s)\varphi(s')$$

**(3)  $\varphi$  is surjective (By construction)**

**(4)  $\ker \varphi = S \cap I$**

If  $s \in S \cap I$ , now  $\varphi(s) = 0 + I$  as  $s \in S \cap I$

$$\Rightarrow S \cap I \subseteq \ker \varphi$$

If  $s \in \ker \varphi$ ,  $\varphi(s) = I$ , now  $s \in S$  and  $s \in I$

$$\Rightarrow s \in S \cap I$$

$$\therefore S \cap I = \ker \varphi$$

By the First Isomorphism Theorem,

$$\frac{S+I}{I} \cong \frac{S}{S \cap I}$$

$\square$

Given:  $D \in \mathbb{Q}$  be a rational number and is NOT a perfect square.

$$R := \{a + b\sqrt{D} \mid a, b \in \mathbb{Q}\}$$

(a)

*Proof.* Observe,  $R \neq \emptyset$  as  $0 \in R$ .

Consider  $x, y \in R$ , where  $x = a_1 + b_1\sqrt{D}$  and  $y = a_2 + b_2\sqrt{D}$ ;  $a_1, b_1, a_2, b_2 \in \mathbb{Q}$ .

$$x - y = (a_1 - a_2) + (b_1 - b_2)\sqrt{D} \in R \text{ as } a_1 - a_2 \in \mathbb{Q} \text{ and } b_1 - b_2 \in \mathbb{Q}.$$

Thus,  $(R, +)$  is a subgroup of  $(\mathbb{C}, +)$ .

Now, look at  $xy = (a_1 + b_1\sqrt{D})(a_2 + b_2\sqrt{D})$

$$\begin{aligned} &= a_1a_2 + b_1b_2D + (a_2b_1 + a_1b_2)\sqrt{D}, \text{ and} \\ & \quad a_1a_2 + b_1b_2D, \quad a_2b_1 + a_1b_2 \in \mathbb{Q}. \end{aligned}$$

Hence,  $xy \in R$  and  $(R, +, \cdot)$  is a subring of  $(\mathbb{C}, +, \cdot)$ . □

(b)

**Theorem 0.6.** Show: If  $(a + b\sqrt{D}) \neq 0$ , then show that  $a^2 - b^2D \neq 0$ .

*Proof.* If  $(a + b\sqrt{D}) \neq 0$  then  $a \neq 0$  and  $b \neq 0$ .

Hence,  $a - b\sqrt{D} \neq 0$ . Now,  $(a + b\sqrt{D})(a - b\sqrt{D}) \neq 0$

$$\Rightarrow a^2 - b^2D + 0(\sqrt{D}) \neq 0.$$

□

(c)

**Theorem 0.7.** Prove:  $R$  is a field.

*Proof.* Observe  $R$  is a commutative, unital ring.

Moreover,  $R$  is a division ring as for  $a + b\sqrt{D} \neq 0$  then  $\frac{a - b\sqrt{D}}{a^2 - b^2D}$  is the inverse with respect to multiplication.

Now,  $R$  is a commutative, division ring, which is a field. □

Given the content in the image, Given  $R$  is a unital ring,

$$Z(CR) := \{a \in R \mid ab = ba \text{ for all } b \in R\}.$$

*Proof.* Observe,  $Z(CR) \neq \emptyset$  as  $1 \in Z(CR)$ .

Now, consider  $a, a' \in Z(CR)$ .

For any  $b \in R$ ,  $(a - a')b = ab - a'b = ba - ba' = b(a - a') \implies (Z(CR), +)$  is a subgroup of  $R$ .

For any  $b \in R$ , observe  $aa'b = aba' = baa' \implies aa' \in Z(CR)$ .

Hence,  $Z(CR)$  is a subring. □